A FORCED COORDINATE WITH NEARLY MAXIMAL ERASURE ENTROPY

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Abstract. The erasure entropy $E_x(f)$ of a Boolean function at a coordinate x is the conditional entropy of the event $x \in U$ given $U \setminus \{x\}$, where U is drawn from the spectral distribution, and it never exceeds the binary entropy of the influence $\mathrm{Inf}_x(f)$. We study an explicit monotone function on $2k+1$ variables in which one coordinate x is the only variable lying both in a shortest minterm and in a shortest maxterm. It is obtained from a majority function with one hidden coordinate by replacing half of the majority's inputs with the OR of two fresh variables and the other half with the AND of two. We prove that $E_x/\mathrm{Inf}_x = (1+o(1))\log_2(1/\mathrm{Inf}_x) = (1+o(1))(k-1)\log_2 \frac{4}{3}$, so the ratio at the forced coordinate grows linearly in k and is asymptotic to its largest possible value. The proof writes the relevant biased Fourier coefficients of the majority as integrals over the unit circle, on which the integrand has an explicit modulus with a single pair of peaks, and uses that the phase advances by $\arccos \frac{1}{3}$, an irrational multiple of π, from one class of coefficients to the next.

1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ and let U be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy-influence conjecture of Friedgut and Kalai [4] asks for an absolute C with $H(U) \le C\,\mathbb{E}|U|$. It is known for symmetric functions and read-once decision trees [11], it is preserved under suitable compositions [10], and weaker forms and bounds with extra factors are in [1, 13, 7, 5]; lower bounds on the best constant are in [6]. The erasure entropy of f at a coordinate x is

$$E_x(f) = H\big(\mathbf{1}[x \in U] \;\big|\; U \setminus \{x\}\big),$$

the conditional entropy of one coordinate given the others, in the sense of Verdú and Weissman [12], and with $\mathrm{Inf}_x(f) = \Pr[x \in U]$ the conjecture holds if and only if

$$\sup_f \min_{x\colon \mathrm{Inf}_x(f)>0} \frac{E_x(f)}{\mathrm{Inf}_x(f)}$$

is finite [16], which builds on the reduction of González, MacManus and Pereyra [3] to a statement about one coordinate; coordinates at which the ratio is bounded for some classes of functions are studied in [17]. Since conditioning does not increase entropy,

$$\frac{E_x(f)}{\mathrm{Inf}_x(f)} \le \frac{h_2(\mathrm{Inf}_x(f))}{\mathrm{Inf}_x(f)} \le \log_2 \frac{e}{\mathrm{Inf}_x(f)} \tag{1}$$

for every f and x with $\mathrm{Inf}_x(f) > 0$.

Which coordinate to take is the difficulty. In [18] several natural selection rules are shown to fail on monotone functions, among them the rule that takes, among the variables lying both in a shortest minterm and in a shortest maxterm, one of smallest ratio. For that rule the counterexample is a family $F_{k,t}$ in which a coordinate x is the only variable of both kinds, and the ratio at x is unbounded when k and t grow together; for fixed t the question was left open. Here we settle the smallest case $t = 1$, which is the simplest member of the family, and find the exact rate of growth.

Fix an odd $k = 2m+1$ and let $y^0 \in \{0,1\}^k$ be the point with $y^0_j = 1$ for $j \le m$ and $y^0_j = 0$ for $j > m$, and $y^1 = y^0 + e_{m+1}$. Let M be the majority function MAJ_k with the point y^1 removed from its true set, and

$$\tilde{f}_k(y,x) = M(y) \vee \big(x \wedge [y \ge y^0]\big).$$

Let F_k be the function of $2k+1$ variables obtained from $\tilde{f}_k$ by replacing y_j with the OR of two fresh variables for $j \le m$ and with the AND of two fresh variables for $j > m$. It is the function $F_{k,1}$ of [18, Section 5]; it is monotone, and x is the only variable that lies both in a shortest minterm and in a shortest maxterm of F_k [18, Section 5].

Theorem 1. $\mathrm{Inf}_x(F_k) = (3/4)^{k-1}$ *and, as* $k \to \infty$ *through odd values,*

$$\frac{E_x(F_k)}{\mathrm{Inf}_x(F_k)} = (1+o(1)) \log_2 \frac{1}{\mathrm{Inf}_x(F_k)} = (1+o(1))\,(k-1) \log_2 \frac{4}{3}.$$

The upper half of the theorem is (1). The exact values computed in Section 5 approach the bound slowly: the quotient of the ratio by $(k-1)\log_2 \frac{4}{3}$ is 0.80 at $k = 21$, 0.88 at $k = 81$ and 0.92 at $k = 141$, increasing in k over the computed range.

2. The ratio as a double sum

All logarithms are to base 2 unless written ln. For $p, q \ge 0$ let $\Phi(p,q) = p \log \frac{p+q}{p} + q \log \frac{p+q}{q}$, with $\Phi(p,0) = \Phi(0,q) = 0$, and $\Psi(r) = \Phi(r^2, 1)$ for $r \ge 0$. Then $\Psi \ge 0$, and $\Psi(r) \ge \log r^2$ for $r \ge 1$, since both terms of Φ are nonnegative and $(r^2+1)/1 \ge r^2$.

Put $P = \frac{3}{4}$ and $q = \frac{1}{4}$. The OR of two uniform bits is true with probability P and the AND with probability q. By the proof of [18, Proposition 12], $E_x(F_k)$ and $\mathrm{Inf}_x(F_k)$ are the erasure entropy and the influence of x for the law $c(S)^2$, where $c(S)$ are the coefficients of $\tilde{f}_k$, in ± 1 values, in the orthonormal basis $\prod_{j \in S} \phi_j(y_j)$ (times $2x-1$ when $x \in S$) of the product measure on (y,x) with $\Pr[y_j = 1] = \mu_j$ and x uniform; here $\mu_j = P$ for $j \le m$, $\mu_j = q$ for $j > m$, and $\phi_j(y) = (y - \mu_j)/\sigma$ with $\sigma = \sqrt{Pq} = \sqrt{3}/4$ (the μ-biased Fourier basis, see [9, Chapter 8]; the conjecture for biased product measures is formulated in [8]).

With $+1$ for true, $\tilde{f}_k = \bar{A} + (2x-1)D$, where

$$D = \mathbf{1}[y = y^0] + \mathbf{1}[y = y^1], \qquad \bar{A} = 2N - 1 + \mathbf{1}[y = y^0] - \mathbf{1}[y = y^1],$$

and $N = \mathbf{1}[|y| \ge m+1]$. For $S \subseteq [k]$ write $u = |S \cap \{1,\dots,m\}|$ and $v = |S \cap \{m+2,\dots,k\}|$, and let $\hat{g}(S)$ denote the coefficient of g at $\prod_{j \in S} \phi_j$. Put

$$X(z) = q + Pz, \qquad Y(z) = P + qz, \qquad H_{uv}(z) = (z-1)^{u+v} X(z)^{m-u} Y(z)^{m+1-v}.$$

Lemma 2. *If* $m+1 \in S$ *then* $\hat{D}(S) = 0$. *If* $m+1 \notin S$ *then*

$$\hat{D}(S) = (-1)^v \sigma^{u+v} P^{2m-u-v}, \qquad \hat{\bar{A}}(S) = 2\sigma^{u+v} t_{uv} - \mathbf{1}[S = \emptyset] + \tfrac{1}{2}\hat{D}(S),$$

where

$$t_{uv} = \sum_{c \ge m+1} [z^c] H_{uv}(z).$$

Proof. For a point w, $\widehat{\mathbf{1}[y=w]}(S) = \Pr[y = w] \prod_{j \in S} \phi_j(w_j)$. For $j \le m$, $y^0_j = 1$ and $\phi_j(1) = q/\sigma = \sqrt{q/P}$; for $j \ge m+2$, $y^0_j = 0$ and $\phi_j(0) = -q/\sigma = -\sqrt{q/P}$. The points y^0 and y^1 differ only at $m+1$, where $\mu_{m+1} = q$, so $\Pr[y = y^0] = P^{2m+1}$ and $\Pr[y = y^1] = P^{2m} q$. If $m+1 \in S$ the two terms of $\hat{D}(S)$ carry the factors $P^{2m+1}\phi_{m+1}(0) = -P^{2m}\sqrt{Pq}$ and $P^{2m} q\, \phi_{m+1}(1) = P^{2m}\sqrt{Pq}$, which cancel. Otherwise both points give the same product $(-1)^v (q/P)^{(u+v)/2}$, so $\hat{D}(S) = P^{2m}(-1)^v (q/P)^{(u+v)/2} = (-1)^v \sigma^{u+v} P^{2m-u-v}$ and $\widehat{\mathbf{1}[y=y^0]}(S) - \widehat{\mathbf{1}[y=y^1]}(S) = (P -$

$q)\hat{D}(S) = \frac{1}{2}\hat{D}(S)$. Finally $\mathbb{E}[z^{y_j}\phi_j(y_j)] = \mu_j z(1-\mu_j)/\sigma - (1-\mu_j)\mu_j/\sigma = \sigma(z-1)$ and $\mathbb{E}[z^{y_j}] = 1-\mu_j+\mu_j z$, so by independence $\mathbb{E}[z^{|y|}\prod_{j\in S}\phi_j(y_j)] = \sigma^{u+v}H_{uv}(z)$ when $m+1 \notin S$, and $\hat{N}(S)$ is the sum of its coefficients at z^c, $c \geq m+1$. □

Let $\beta_u = \binom{m}{u}q^u P^{m-u}$ be the binomial law with parameters m and q, and for $0 \leq u, v \leq m$ put

$$r_{uv} = \frac{\left|2t_{uv} - \mathbf{1}[u=v=0] + \frac{1}{2}(-1)^v P^{2m-u-v}\right|}{P^{2m-u-v}}.$$

Lemma 3. $\mathrm{Inf}_x(F_k) = P^{2m} = (3/4)^{k-1}$ *and*

$$\frac{E_x(F_k)}{\mathrm{Inf}_x(F_k)} = \sum_{u,v=0}^{m} \beta_u\beta_v\,\Psi(r_{uv}).$$

Proof. By the reduction above, $E_x = \sum_S \Phi(\hat{\tilde{A}}(S)^2, \hat{D}(S)^2)$ and $\mathrm{Inf}_x = \sum_S \hat{D}(S)^2$, the sums over $S \subseteq [k]$; the terms with $m+1 \in S$ vanish. There are $\binom{m}{u}\binom{m}{v}$ sets with given u, v and $m+1 \notin S$, and $\sum_{u,v}\binom{m}{u}\binom{m}{v}(Pq)^{u+v}P^{2(2m-u-v)} = (Pq+P^2)^{2m} = P^{2m}$. By homogeneity $\Phi(\hat{\tilde{A}}^2, \hat{D}^2) = \hat{D}^2\,\Psi(|\hat{\tilde{A}}|/|\hat{D}|)$, the quotient $|\hat{\tilde{A}}(S)|/|\hat{D}(S)|$ is r_{uv} by Lemma 2, and $\binom{m}{u}\binom{m}{v}(Pq)^{u+v}P^{2(2m-u-v)}/P^{2m} = \beta_u\beta_v$. □

So E_x/Inf_x is the mean of $\Psi(r_{uv})$ when u and v are independent binomial variables with parameters m and $\frac{1}{4}$. Most of that law sits near $u = v = m/4$, and the theorem says that there $\log r_{uv}^2$ is close to $2m\log\frac{4}{3}$ for almost all (u, v). The difficulty is that t_{uv} is a sum of terms of both signs; we show that it oscillates, in u, like the cosine of an irrational rotation times an exponentially large envelope.

3. The tail on the unit circle

Lemma 4. *If $u + v \geq 1$ then $Q_{uv} = H_{uv}/(z-1)$ is a polynomial and*

$$t_{uv} = [z^m]\,Q_{uv}(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} Q_{uv}(e^{\mathrm{i}\theta})\,e^{-\mathrm{i}m\theta}\,d\theta.$$

Proof. Write $Q_{uv} = \sum_c g_c z^c$. Then $[z^c]H_{uv} = g_{c-1} - g_c$, and the sum over $c \geq m+1$ telescopes to g_m because Q_{uv} has degree one less than H_{uv}. The integral is Cauchy's formula on $|z| = 1$; coefficient extraction of this kind, followed by a Laplace or saddle-point estimate, is standard [2, Chapter VIII]. □

On the circle the three factors of Q_{uv} have explicit moduli. For $z = e^{\mathrm{i}\theta}$ and $w = \cos\theta$,

$$|z-1|^2 = 2-2w, \qquad |X(z)|^2 = |Y(z)|^2 = q^2 + P^2 + 2Pqw = \frac{5+3w}{8}.$$

Put $n_1 = u+v-1$ and $n_2 = 2m+1-u-v$, so that $n_1+n_2 = 2m$. Then $|Q_{uv}(e^{\mathrm{i}\theta})| = e^{\rho(\cos\theta)}$ with

$$\rho(w) = \frac{n_1}{2}\ln(2-2w) + \frac{n_2}{2}\ln\frac{5+3w}{8}, \qquad \rho''(w) = -\frac{n_1}{2(1-w)^2} - \frac{9n_2}{2(5+3w)^2} \leq -\frac{9n_2}{128}.$$

So ρ is strictly concave on $[-1, 1)$; its maximum is at $w_0 = (3n_2 - 5n_1)/(6m)$ when this lies in $(-1, 1)$, and a short computation gives $2 - 2w_0 = 8n_1/(3m)$, $(5+3w_0)/8 = n_2/(2m)$, hence

$$\rho(w_0) = \frac{n_1}{2}\ln\frac{8n_1}{3m} + \frac{n_2}{2}\ln\frac{n_2}{2m}. \tag{2}$$

At $u = v = m/4$ one has $n_1 \approx m/2$, $w_0 \approx \frac{1}{3}$ and $e^{\rho(w_0)} \approx (\sqrt{3}/2)^m$.

For $\lambda \geq 1$ let W_λ be the set of (u,v) with $|u - m/4| \leq \lambda\sqrt{m}$ and $|v - m/4| \leq \lambda\sqrt{m}$. In what follows c, C denote positive constants that may depend on λ but not on m or on $(u,v) \in W_\lambda$, and m is large enough for each statement. On W_λ, $n_1 = m/2 + O(\sqrt{m})$ and $w_0 = \frac{1}{3} + O(m^{-1/2})$; put $\theta_0 = \arccos w_0$, $\theta^* = \arccos \frac{1}{3}$ and $z_0 = e^{\mathrm{i}\theta_0}$.

Lemma 5. *On W_λ, $\rho(w_0) + (n_2 - 1)\ln\frac{4}{3} \geq (m-1)\ln\frac{4}{3}$.*

Proof. Regard the left side, by (2), as a function $G(n_1)$ of a real variable $n_1 \in (0, 2m)$, with $n_2 = 2m - n_1$. Then $G'(n_1) = \frac{1}{2}\ln\frac{16n_1}{3n_2} - \ln\frac{4}{3}$, which vanishes at $n_1 = m/2$, and $G''(n_1) = \frac{1}{2n_1} + \frac{1}{2n_2} > 0$. So G is convex with its minimum at $m/2$, and $G(m/2) = \frac{m}{4}\ln\frac{4}{3} + \frac{3m}{4}\ln\frac{3}{4} + (\frac{3m}{2} - 1)\ln\frac{4}{3} = (m-1)\ln\frac{4}{3}$. □

Fix $(u,v) \in W_\lambda$ and let $\Lambda(\theta) = \ln Q_{uv}(e^{\mathrm{i}\theta}) - \mathrm{i}m\theta$, for complex θ with $|\theta - \theta^*| \leq \delta$, where $\delta > 0$ is so small that $e^{\mathrm{i}\theta}$ stays away from 1, $-\frac{1}{3}$ and -3, the zeros of $z-1$, X and Y; the logarithm is the branch continuous in this disc. Since Λ is a sum of at most $2m+1$ logarithms of functions analytic and nonvanishing there, plus a linear term, $|\Lambda''|$ and $|\Lambda'''|$ are at most Cm on the disc. For real θ, $\operatorname{Re}\Lambda(\theta) = \rho(\cos\theta)$, so $\operatorname{Re}\Lambda'(\theta_0) = -\rho'(w_0)\sin\theta_0 = 0$ and $\operatorname{Re}\Lambda''(\theta_0) = \rho''(w_0)\sin^2\theta_0 \leq -cm$. Put

$$b = \operatorname{Im}\Lambda'(\theta_0), \qquad A = -\Lambda''(\theta_0), \qquad T_{uv} = \frac{1}{\sqrt{2\pi A}}\exp\Big(\Lambda(\theta_0) - \frac{b^2}{2A}\Big),$$

with the principal square root ($\operatorname{Re} A > 0$).

Lemma 6. *On W_λ: $|b| \leq C\sqrt{m}$, $|T_{uv}| \geq c\,m^{-1/2}e^{\rho(w_0)}$, and $t_{uv} = 2\operatorname{Re}T_{uv} + O(|T_{uv}|\,m^{-1/5})$.*

Proof. Write $a = u/m$, $a' = v/m$. Then $\Lambda'(\theta)/m$ differs by $O(1/m)$ from $\mathrm{i}z\big(\frac{a+a'}{z-1} + \frac{(1-a)P}{X} + \frac{(1-a')q}{Y}\big) - \mathrm{i}$ at $z = e^{\mathrm{i}\theta}$, which is a smooth function of (a, a', θ) and vanishes at $a = a' = \frac{1}{4}$, $\theta = \theta^*$: there $z^* = e^{\mathrm{i}\theta^*} = \frac{1}{3} + \frac{2\sqrt{2}}{3}\mathrm{i}$, and $\frac{1}{2(z^*-1)} + \frac{9}{16X(z^*)} + \frac{3}{16Y(z^*)} = \frac{1}{z^*}$, as one checks from $X(z^*) = \frac{1}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$ and $Y(z^*) = \frac{5}{6} + \frac{\sqrt{2}}{6}\mathrm{i}$. Since $a - \frac{1}{4}$, $a' - \frac{1}{4}$ and $\theta_0 - \theta^*$ are $O(m^{-1/2})$, $|\Lambda'(\theta_0)| \leq C\sqrt{m}$, which bounds b. Next, $\operatorname{Re}(b^2/A) = b^2\operatorname{Re}A/|A|^2 \leq C$, so $|T_{uv}| \geq c\,|A|^{-1/2}|e^{\Lambda(\theta_0)}| \geq c\,m^{-1/2}e^{\rho(w_0)}$.

By Lemma 4, $t_{uv} = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\Lambda(\theta)}d\theta$, the integrand being $Q_{uv}(e^{\mathrm{i}\theta})e^{-\mathrm{i}m\theta}$ on the whole circle. Split the interval into $J_+ = \{|\theta - \theta_0| \leq m^{-2/5}\}$, $J_- = -J_+$ and the rest. On J_+, with $\theta = \theta_0 + h$, $\Lambda(\theta) = \Lambda(\theta_0) + \mathrm{i}bh - \frac{1}{2}Ah^2 + R(h)$ with $|R(h)| \leq Cm|h|^3 \leq Cm^{-1/5}$. Hence

$$\frac{1}{2\pi}\int_{J_+} e^{\Lambda} = \frac{e^{\Lambda(\theta_0)}}{2\pi}\Big(\int_{\mathbb{R}} e^{\mathrm{i}bh - Ah^2/2}dh - \int_{|h|>m^{-2/5}} e^{\mathrm{i}bh - Ah^2/2}dh\Big) + O\Big(e^{\rho(w_0)}m^{-1/5}\int_{\mathbb{R}} e^{-cmh^2}dh\Big).$$

The first integral is $\sqrt{2\pi/A}\,e^{-b^2/(2A)}$, which makes the first term T_{uv}; the second is at most $Ce^{-cm^{1/5}}$; the error term is $O(e^{\rho(w_0)}m^{-1/2-1/5}) = O(|T_{uv}|m^{-1/5})$. The coefficients of Q_{uv} are real, so the integral over J_- is the complex conjugate. Off $J_+ \cup J_-$, for $\theta \in [0,\pi]$, $|\cos\theta - \cos\theta_0| = 2\sin\frac{\theta+\theta_0}{2}\big|\sin\frac{\theta-\theta_0}{2}\big| \geq c|\theta - \theta_0|$ because θ_0 stays away from 0 and π, and likewise for $\theta \in [-\pi, 0]$; since $\rho'(w_0) = 0$ and $\rho'' \leq -9n_2/128 \leq -cm$, the integrand there has modulus at most $e^{\rho(w_0) - cm(\cos\theta - w_0)^2} \leq e^{\rho(w_0) - cm^{1/5}}$, which is $O(|T_{uv}|m^{-1/5})$ after integration. □

Lemma 7. *Let $\alpha = \arccos\frac{1}{3}$. If (u,v) and $(u+1,v)$ lie in W_λ, then $\arg T_{u+1,v} - \arg T_{uv} = \alpha + O(m^{-1/2})$ modulo 2π.*

Proof. Let $g(z) = (z-1)/X(z)$, so that $Q_{u+1,v} = gQ_{uv}$, and denote by Λ^+, θ_0^+, b^+, A^+ the quantities for $(u+1, v)$. Here $\Lambda^+ = \Lambda + \ln g(e^{\mathrm{i}\theta})$, and w_0 changes by $-4/(3m)$, so $\theta_0^+ - \theta_0 = O(1/m)$. Hence

$$\Lambda^+(\theta_0^+) - \Lambda(\theta_0) = \ln g(z_0) + \Lambda'(\theta_0)(\theta_0^+ - \theta_0) + O(1/m) = \ln g(z_0) + O(m^{-1/2})$$

by Lemma 6. Similarly $A^+ - A = O(1)$ and $b^+ - b = O(1)$, so $\arg A$ moves by $O(1/m)$ and $b^2/(2A)$ by $O(|b|/m + b^2/m^2) = O(m^{-1/2})$. So $\arg T$ moves by $\operatorname{Im} \ln g(z_0) + O(m^{-1/2})$. Finally $g(z^*) = \frac{4}{3}z^*$, as $(z^*-1)\overline{X(z^*)} = \frac{1}{3} + \frac{2\sqrt{2}}{3}\mathrm{i}$ and $|X(z^*)|^2 = \frac{3}{4}$; thus $\arg g(z^*) = \theta^* = \alpha$, and $z_0 - z^* = O(m^{-1/2})$. □

The number α/π is irrational: by Niven's theorem [15], if α/π is rational and $\cos\alpha$ is rational then $\cos\alpha \in \{0, \pm\frac{1}{2}, \pm 1\}$.

4. Proof of the theorem

The equality $\mathrm{Inf}_x(F_k) = (3/4)^{k-1}$ is Lemma 3, and the upper bound is (1). For the lower bound fix $\varepsilon \in (0, \frac{1}{4})$, $\lambda \geq 1$ and an integer $L \geq 1$. Since α/π is irrational, the sequence $\ell\alpha/\pi$ is equidistributed modulo 1 [14], with discrepancy tending to 0; the set of ψ modulo π with $|\cos\psi| < 2\varepsilon$ is an interval of length $2\arcsin 2\varepsilon \leq 2\pi\varepsilon$. So we may take L so large that for every real ψ_0 at most $3\varepsilon L$ of the numbers $\ell = 0, \dots, L-1$ satisfy $|\cos(\psi_0 + \ell\alpha)| < 2\varepsilon$.

Fix v and cut the set of u with $(u,v) \in W_\lambda$ into consecutive blocks of L values, leaving at most $L-1$ values over. In a block starting at u_0, Lemma 7 gives $\arg T_{u_0+\ell,v} = \arg T_{u_0,v} + \ell\alpha + O(Lm^{-1/2})$ for $\ell < L$, and for m large the error is below ε, so at most $3\varepsilon L$ values of u in the block have $|\cos\arg T_{uv}| < \varepsilon$. On W_λ, $\beta_{u+1}/\beta_u = (m-u)/(3(u+1)) = 1 + O(m^{-1/2})$, so the weights β_u in a block differ by a factor at most 2 for m large, and the bad values carry at most 6ε of the block's weight. The left-over values carry weight at most $(L-1)\max_u \beta_u = O(Lm^{-1/2})$ in each row. Summing over v, the pairs $(u,v) \in W_\lambda$ with $|\cos\arg T_{uv}| < \varepsilon$, together with the left-over ones, have $\beta_u\beta_v$-weight at most $6\varepsilon + o(1)$.

Let $(u,v) \in W_\lambda$ have $|\cos\arg T_{uv}| \geq \varepsilon$. By Lemma 6, $|t_{uv}| \geq 2\varepsilon|T_{uv}| - C|T_{uv}|m^{-1/5} \geq \varepsilon|T_{uv}| \geq c\varepsilon m^{-1/2} e^{\rho(w_0)}$, and $(u,v) \neq (0,0)$. With $2m - u - v = n_2 - 1$ and Lemma 5,

$$r_{uv} \geq \frac{2|t_{uv}|}{P^{n_2-1}} - \frac{1}{2} \geq c\varepsilon m^{-1/2}\Big(\frac{4}{3}\Big)^m - \frac{1}{2},$$

which exceeds 1 for m large, and then $\Psi(r_{uv}) \geq \log r_{uv}^2 \geq 2m\log\frac{4}{3} - \log m - C_\varepsilon$. By Chebyshev's inequality $\Pr[|u - m/4| > \lambda\sqrt{m}] \leq \frac{3}{16\lambda^2}$, so W_λ has weight at least $(1 - \frac{3}{16\lambda^2})^2$. Since $\Psi \geq 0$, Lemma 3 gives

$$\frac{E_x(F_k)}{\mathrm{Inf}_x(F_k)} \geq \Big(\big(1 - \tfrac{3}{16\lambda^2}\big)^2 - 6\varepsilon - o(1)\Big)\Big(2m\log\frac{4}{3} - \log m - C_\varepsilon\Big).$$

As $2m = k-1$, dividing by $(k-1)\log\frac{4}{3}$ and letting $m \to \infty$, then $\lambda \to \infty$ and $\varepsilon \to 0$, gives $\liminf \geq 1$. □

Remark 8. In v the factor is $(z-1)/Y$, and $(z^*-1)/Y(z^*) = \frac{4}{3}(-\frac{1}{3} + \frac{2\sqrt{2}}{3}\mathrm{i})$, so the phase step is $\pi - \alpha$ and the step from (u,v) to $(u+1, v+1)$ is π: to first order the sign of t_{uv} alternates along diagonals.

5. Numerical checks

Lemma 3 turns the ratio into a double sum of $(m+1)^2$ terms whose ingredients are integers over powers of 4, so the coefficients can be computed exactly in rational arithmetic; each term of Φ is then evaluated as $p\log(1 + q/p)$ with a $\log(1+\cdot)$ routine, since p/q reaches 10^{17} for k above 120. The ratio $R_k = E_x(F_k)/\mathrm{Inf}_x(F_k)$ and the quotient $R_k/((k-1)\log\frac{4}{3})$ are:

k	R_k	quotient	k	R_k	quotient
21	6.631	0.799	101	37.249	0.898
41	13.826	0.833	121	45.276	0.909
61	21.448	0.861	133	50.138	0.915
81	29.311	0.883	141	53.319	0.918

For $k = 3, 5, 7, 9$ the values 1.0512, 1.8769, 2.5690, 3.1179 agree to 10^{-9} with the ratio computed from the full Walsh–Hadamard transform of F_k on 7, 11, 15 and 19 variables, which also confirms that x is the only variable in both a shortest minterm and a shortest maxterm. The Laplace approximation of Lemma 6 was compared with the exact t_{uv} over all of W_1: the largest error relative to $|T_{uv}|$ is 0.149, 0.095 and 0.071 at $k = 41$, 81 and 141, about $0.6\,m^{-1/2}$. A scan of $|Q_{uv}(e^{i\theta})|$ for $n_1/m \in [0.4, 0.6]$ finds no local maximum on $(0, \pi)$ other than θ_0.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [21] and Math Brain [19]. The family $F_{k,t}$ is from [18]. Claude Opus 5.5 (Anthropic) found Theorem 1 and its proof, wrote the numerical checks and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-forced-coordinate`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [19].

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